

Research Methods

Day 2: From Curiosity to Research Questions

Outline for Today

1. **Team Work & Day 1 Refresher**
2. **Your Research Question is a Compass**
3. **What Makes a Good Question? BRIC-FOR**
4. **Narrowing Topics with the 5 W's (and How)**

Welcome Back! Day 2

Let's start with a teamwork & a refresher.

Then we shall learn the very first step of the research process: **asking a good question!**





TEAM WORK

Team Work

-  Let's review your homework with your teammates.
-  Share your curiosities and thoughts regarding your chosen global issue.

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-  Share your curiosities and thoughts regarding your chosen global issue.

Find common themes. Identify favorite ideas. Select 1-2 questions to refine later.



Let's Refresh Our Memories: What is Research?



A Journey of Discovery

Research is a systematic investigation to establish facts and reach new conclusions.



Question-Driven

Every great discovery begins with a well-framed question that guides your direction.

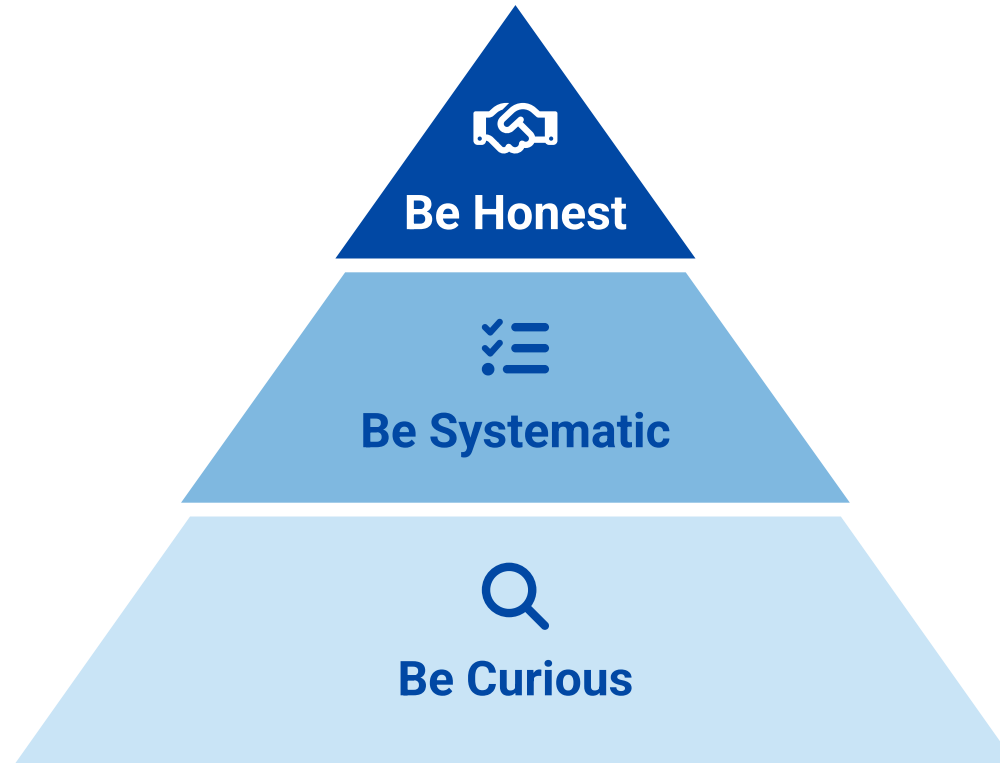


Evidence-Based

Research relies on gathering and analyzing evidence rather than opinions or assumptions.

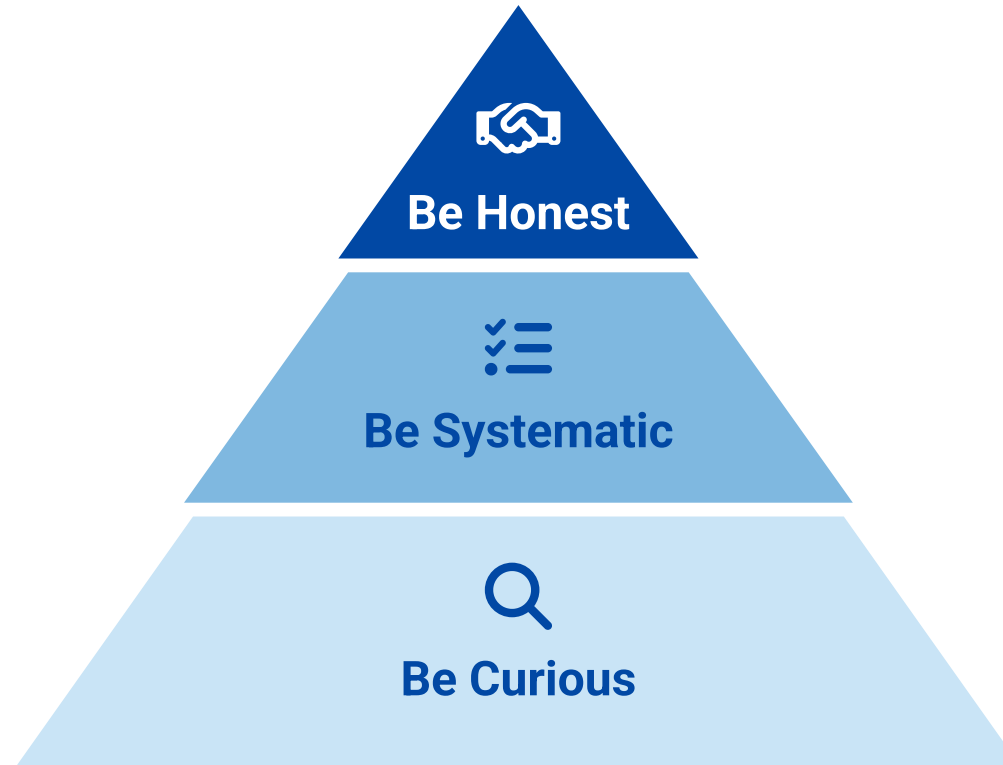
Refresher cont'd: Core Principles of Good Research

No matter what the field is



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Research begins with curiosity.

Refresher cont'd: Principle 1 – Be Curious

The Starting Point: Curiosity

Wonder — Asking questions such as

"Why?" "When?" "How much?"

"What if?" "How come?" "For whom?"

"Under what circumstances?"

"What is the trend?"

"What is the relationship between?"

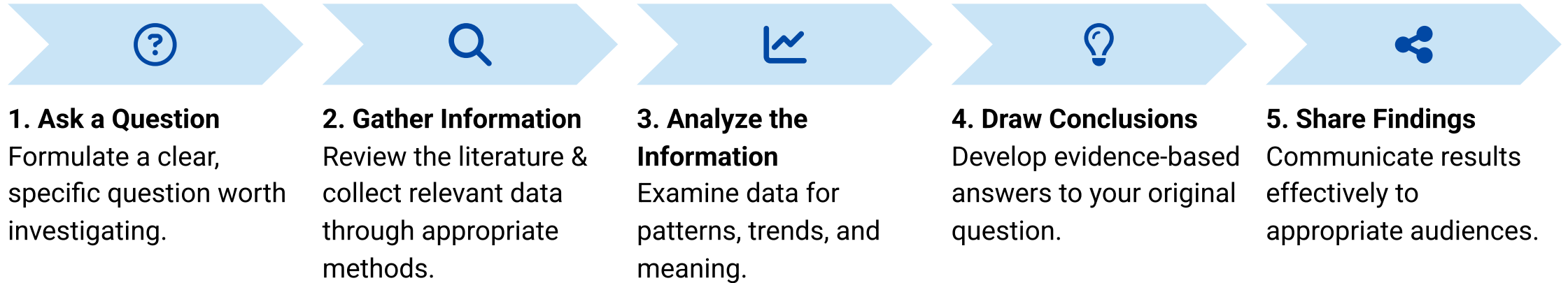
drives meaningful research.



Research begins with curiosity and asking meaningful questions.

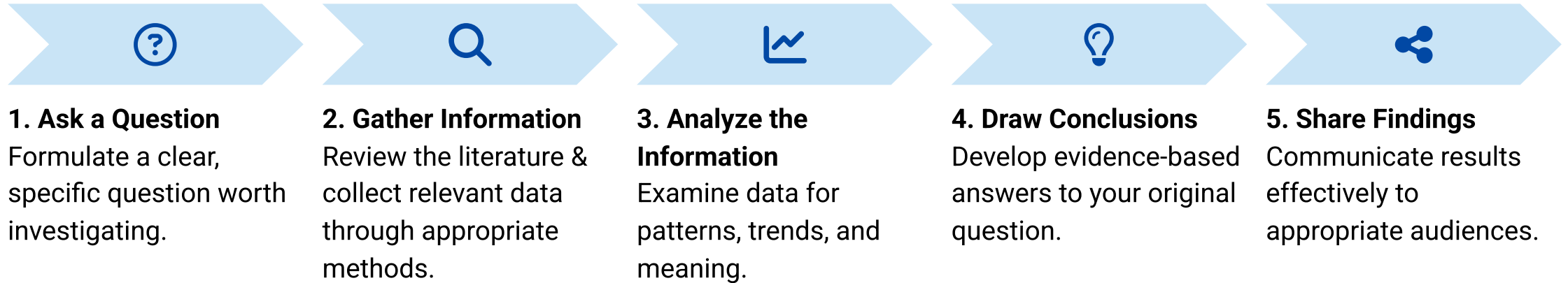
Refresher cont'd: The Research Process

The journey from question to insight follows these essential steps:



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The journey from question to insight follows these essential steps:



Today we focus on Step 1: Asking a Good Question!

Your Research Question is a Compass



📍 **A well-framed question acts as your compass, pointing you toward useful answers.**

🔄 **Vague questions lead to circular research. Specific questions drive you to meaningful discoveries.**

⚠️ **Without the right question, even the best research efforts might not yield satisfying answers.**

Case Study: The Library Problem

1

Initial Question

"How can we make teens read more books?"

2

Reframed Question

"What are the barriers to teens' library use?"

3

Research Method

Surveyed classmates about library usage.

4

Discovery

Limited Wi-Fi and few comfortable spaces deterred teens.



Case Study: Solar Panel Investigation

Initial Question

"How can we improve solar panel efficiency?"

Better Question

"What factors are limiting our school's solar panel performance?"

Research Findings

Shade, incorrect tilt, and dust accumulation were the issues.

What Makes a Good Research Question? BRIC-FOR



Balanced (Well-Scoped)

Rellevant & **I**nteresting

Clear & **F**ocused

Open-ended

Rearchable

BRIC-FOR: Balanced in Scope (aka Well-Scoped Questions)

Too Broad

"What is the history of the United States?"

Spans hundreds of years and countless topics.

Too Narrow

"What was the price of bread in our town on January 1, 1950?"

More of a trivia fact than a research project.

Just Right

"How have food prices in our town changed from 1950 to today, and why?"

Focused yet allows for meaningful analysis.

BRIC-FOR: Relevant and Interesting Questions

💖 Personal Relevance

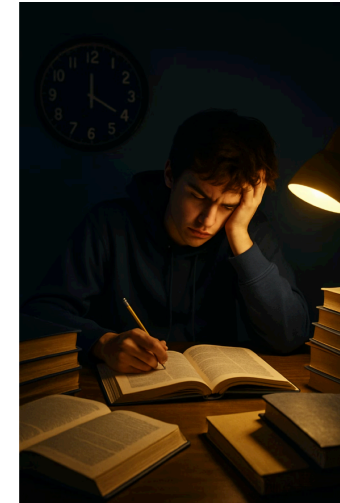
Choose topics that matter to you—you'll stay more engaged.

👥 Community Impact

Research that addresses local issues can have real-world applications.

💡 Example

"How does sleep affect test scores among students in my grade?"



BRIC-FOR: Clear and Focused Questions

Too Broad

"What causes climate change?"

This spans science, policy, history, and economics—a lifetime of work!

Well-Focused

"How do car emissions contribute to climate change in my city?"

Narrows to one cause, one effect, and one location.

BRIC-FOR: Open-Ended Questions

Closed-Ended

"Do video games affect mood?"

Can be answered with just "yes" or "no"—limited exploration.

- Easy to quantify and analyze
- Good for specific, categorizable information such as "Do you have a library card?"
- Useful for gathering straightforward facts or opinions you can count

Open-Ended

"How might playing video games affect the mood of middle school students?"

Invites exploration of different effects, game types, and conditions.

- Require detailed, free-form responses
- Provide rich, nuanced information
- Harder to analyze quickly

BRIC-FOR: Researchable Questions

Not Easily Researchable

"Is there life on other planets?"

Fascinating but beyond a student's ability to gather original data.

Researchable Version

"What do high school students believe about life on other planets?"

Can be investigated through surveys and interviews.

Key Consideration

Can you gather information to answer this using available tools?





Let's Play a Game: WEAK or STRONG Research Questions?

"Is pollution bad?"

"Is pollution bad?"

✗ WEAK!

Vague, obvious, closed-ended and not specific about type of pollution or effects.

"What effects does air pollution have on respiratory health in children?"

"What effects does air pollution have on respiratory health in children?"

✓ **STRONG!**

Specifies pollution type, health aspect, and affected population.

"How do we fix the education system?"

"How do we fix the education system?"

✗ WEAK!

Too broad, assumes problems, and implies one solution exists.

"What challenges do students face in remote learning, and how do these affect performance?"

"What challenges do students face in remote learning, and how do these affect performance?"

✓ **STRONG!**

Focused on specific context with clear variables to examine.



Narrowing Down a Topic



Start with a broad interest — Example: "Climate change"—a general theme



Find a niche or sub-topic — Example: "Effects of rising sea levels on coastal communities"



Apply the 5 W's (and How) — Who, What, Where, When, Why, and How



Turn it into a specific question — Example: "How are rising sea levels affecting fishing communities in Florida?"

Using the 5 W's (and How)



Who

Who is involved or affected?
(teenagers, athletes, teachers,
polar bears)



What

What aspect interests you?
(mental health, economic cost,
performance)



Where

A certain location or context?
(your city, online communities,
lakes)



When

A time period? (past 5 years, during summer,
historical period)



Why/How

Causes or processes? (why something happens,
how factors interact)

Example: Narrowing "Video Games"

Broad Topic

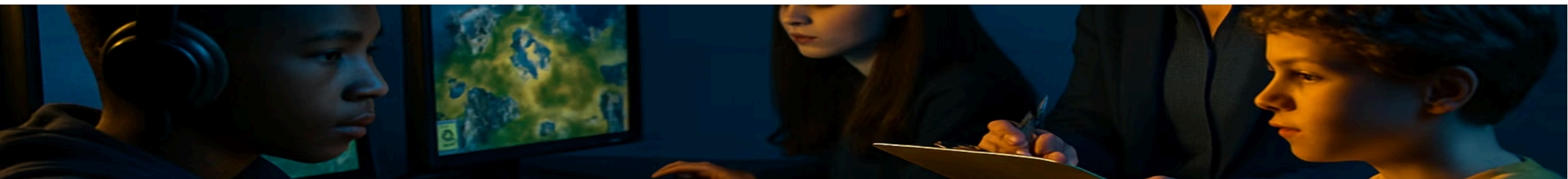
Video games (huge topic with many aspects)

Narrower Focus

Video games and learning (educational aspect)

Specific Angle

Video games and problem-solving skills



From Topic to Research Question

Broad Topic

Video games

Research Question

"Can playing video games improve middle school students' problem-solving skills, and if so, how?"

Narrower Focus

Video games and problem-solving skills

Elements

- **Who:** middle school students
- **What:** effect on problem-solving skills
- **Open-ended:** asks "how" not just "if"



YOUR TURN!

YOUR TURN! Let's Team UP to merge & refine your potential research questions

Make sure to use the 5 W's (& H)



Who

Who is involved or affected? (teenagers, athletes, teachers, polar bears)



What

What aspect interests you? (mental health, economic cost, performance)



Where

A certain location or context? (your city, online communities, lakes)



When

A time period? (past 5 years, during summer, historical period)



Why/How

Causes or processes? (why something happens, how factors interact)

Team Homework: Handout 2

From curiosity to formulating research questions

- 💡 **What you know/assume & want to learn**
- 1/2 **3-5 narrowed down research questions**
- ☑ **Evaluate them using BRIC-FOR**



Homework



**WHAT WE KNOW
ABOUT THIS ISSUE**



**WHAT WE WANT TO
KNOW**

Homework: Come up with 3-5 potential research questions

RATE each question using **BRIC-FOR** criteria

Balanced spectrum? Relevant? Interesting? Clear? Focused? Open-ended? Researchable?

1

**Not so
much**

2

3

Somewhat

4

5

**Very
much**

EXTRA SLIDES FOR DAY 2



Let's Play a Game!

Let's take your research questions to the next level.

-  Each team will receive a card with a "**constraint**".
-  Update your question following the hint on the card.

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-  Each team will receive a card with a "**constraint**".
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Now it is time to present your question to the larger group and get rated on **BRIC-FOR**

Balanced spectrum? Relevant? Interesting? Clear? Focused? Open-ended? Researchable?

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